Q**1. What is a file? Explain the need of file in Python.**

Ans:

A file is a container used to store data permanently on a computer. In Python, files are used to store and retrieve information whenever required.

Need of files in Python:

1. Permanent Storage: Files store data permanently, even after the program is closed.
2. Data Retrieval: We can read previously stored data whenever needed.
3. Data Sharing: Files allow us to share information between different programs.
4. Handling Large Data: Files help us manage large amounts of information easily.
5. Record Keeping: We can store records like student details, employee information and sales data.

Example:

```
file = open("student.txt", "w")

file.write("Apurva")

file.close()
```

This program creates a file named student.txt and stores the name Apurva in it.

Q2. State and explain the different modes of opening a file in Python.

Ans:

Python provides different file modes to perform operations such as reading, writing and appending data.

Mode	**Name**	**Explanation**
r	Read	Opens a file for reading. Gives an error if the file does not exist.
w	Write	Opens a file for writing. Creates a new file or overwrites existing content.
a	Append	Adds new content at the end of an existing file without deleting previous content.
x	Exclusive creation	Creates a new file. Gives an error if the file already exists.
r+	Read and Write	Opens a file for both reading and writing.
w+	Write and Read	Opens a file for reading and writing, but overwrites existing content.

a+	Append and Read	Opens a file for reading and appending new content.

Other file mode options:

- t – Text mode (default mode).
- b – Binary mode (used for images, videos and other binary files).

Q3. State the difference between Write and Append mode in opening a file in Python.

Ans:

Write mode (w) and Append mode (a) are used to write data into a file, but they work differently.

Write Mode (w)	**Append Mode (a)**
Used to write data into a file.	Used to add new data to an existing file.
Deletes previous content before writing.	Keeps previous content safe.
Creates a new file if it does not exist.	Also creates a new file if it does not exist.
Writes data from the beginning.	Adds data at the end of the file.

Example of Write Mode:

```
f = open("demo.txt", "w")
f.write("Hello")
f.close()
```

Output in file:

Hello

If the file already contained Welcome, that old content is deleted.

Example of Append Mode:

```
f = open("demo.txt", "a")
f.write(" Python")
f.close()
```

Output in file:

Hello Python

The previous content remains, and new content is added at the end.